AYUSH SHARMA

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EDUCATION

Northeastern University (Khoury College of Computer Sciences) 09/2024 - 08/2026
M.S. Artificial Intelligence, GPA: 3.87/4.0 Boston, MA

B.M.S. College of Engineering 08/2020 - 08/2024
B.E. Artificial Intelligence and Machine Learning, GPA: 3.82/4.0 Bangalore, India
Courses: Probability, Statistics, Big Data Analytics, Software Development, Design Patterns, Cloud Computing, Data Structures, OOP

TECHNICAL SKILLS

Programming Languages and Frameworks: Python, C, C++, Java, SQL, Flask, FastAPI, React, REST
Packages: NumPy, Pandas, TensorFlow, PyTorch, Sklearn, PySpark, XGBoost, NLTK, SpaCy, Pytest, Unittest, Pydantic, Hugging Face
Tools: Azure, AWS, FAISS, Neo4j, Git, Docker, Kubernetes, LangChain, LangGraph, LlamaIndex, MLflow, CrewAI, Redis, Kafka, Linux
Core Skills: Machine Learning, Data Science, Backend Engineering, RAG Systems, Agentic AI, Post Training, LLM Fine-Tuning, Testing

WORK EXPERIENCE

Johnson & Johnson 07/2025 - 12/2025
Data Scientist Intern - Applied AI Research, Algorithms Engineering *Boston, MA, USA*

- Performed **supervised fine-tuning** of a **Transformer**-based clinical **NER** model **(F1 > 0.9)**, benchmarking **PEFT** strategies (**LoRA, adapter tuning**) to reduce model memory footprint by **67%** while preserving accuracy across held-out clinical texts.
- Defined span-level evaluation protocols (**nervaluate**) to measure model performance and diagnose boundary and misclassification errors across large clinical datasets, improving extraction reliability. Curated a **gold dataset** for clinical NER benchmarking.
- Designed an LLM-orchestrated **agentic workflow** on **Azure ML** to parse, classify, and extract features from clinical documents ingested via **Azure Blob Storage**, automating a previously manual workflow. Focused on **inference optimization** via **prompt caching**.
- Integrated an **LLM-as-a-judge** self-eval loop to validate outputs and trigger corrective regeneration, reducing manual effort by ∼**90%**.
- Researched methods to measure cardiac output using **time series sensor data** (hemodynamics).

Honeywell 03/2024 - 07/2024
Software Engineer Intern - Operational Intelligence *Bangalore, India*

- Built a **digital twin** KPI optimization workflow, ingesting warehouse data from **Databricks** to generate simulated workforce configurations via predictive modeling and genetic algorithms to improve operational efficiency, delivering **20%** gains in throughput.
- Designed a **closed-loop evaluation** layer benchmarking digital twin outputs against core warehouse KPIs sourced from Databricks, compressing simulation-to-decision time from **24 hours to under 3 hours**. Contributed to a **U.S. Patent filing**.

Purple Advisors 10/2023 - 01/2024
Software Engineer Intern *San Francisco, CA, USA (Remote)*

- Predicting Prospective Customer Engagement: Built a **tree-based classifier** to predict B2B prospect conversion from industry, revenue, and business-model features, surfacing high-likelihood leads and customers for sales. Achieved **84%** accuracy on held-out test data.
- Prompt engineered **GPT-3.5-Turbo** to automate personalized marketing outreach tailored to each prospect's industry and business profile. Performed **A/B testing** against baseline outreach, improving prospect response rates by **30%** over baseline.

Bosch 08/2023 - 10/2023
Software Engineer Intern *Bangalore, India*

- Delivered a **data-driven inventory optimization ETL pipeline** generating safety stock **recommendations** using statistical demand variability and inventory signals, achieving **98%+** service reliability and improving planning accuracy by **15%**.
- Reduced computation time from **48 hours to 1 hour** and saved over 200 manual hours annually for supply chain customers and teams.

PROJECTS

FlowMind: Verify-and-Repair Multi-Agent Flowchart Understanding and Code Synthesis 05/2026 - 08/2026

- Developed a **multi-agent** system for flowchart understanding and code synthesis, combining fine-tuned **VLM** for image extraction, deterministic graph-based structural reasoning, and LLM-based query routing, code generation, and **evaluation harness**.

MarketSensAI (Master's Capstone Project) 01/2026 - 04/2026

- Built a **multi-agent** financial analytics system using **LangGraph** and **MCP**, coordinating agents for news analysis, story clustering, and price event attribution to generate investor reports explaining equity price movements, reducing research time by over **95%**.

Research Buddy (An Informed Research Assistant) 09/2024 - 12/2024

- Engineered a **Graph RAG** system using **LlamaIndex** and **Neo4j** for context retrieval across **knowledge graphs**, integrated **Qwen2.5** for user query handling, achieving **30 ms avg. ITL** and **3-5s** end-to-end query latency with improved retrieval precision.

Physics-Informed Neural Network (PINN) Framework for Solving PDEs 06/2023 - 06/2024

- Led a research initiative using PINNs with custom **physics-informed loss functions** to solve higher-order fluid mechanics differential equations, achieving convergence losses of **1e-6 to 1e-10** and reducing compute time by **40%** vs. Finite Element methods.

PATENT & PUBLICATIONS

- **U.S. Patent filed:** Machine Learning Based Predictive Workload Model to Validate Workforce Key Performance Indicators.
- **Journal Publication:** Stagnation Point Flow of Sutterby Fluid over a Stretching Sheet with an Inclined Magnetic Field: Gradient Descent Method, *International Journal of Modelling and Simulation* (Taylor & Francis), **Q1 journal**.
- **Publications:** 5 Peer-Reviewed IEEE conference papers (ICWITE '26, ICCSAI '25, INCET '23, GCAT '23) on NLP, Transfer Learning, Multimodal AI, Automated Reasoning, Workflow Automation, Computer Vision.

EXTRACURRICULARS

- **Teaching Assistant** for CS6180 - **Generative AI**, Khoury College, Northeastern University under Dr. Mina Ghashami.
- **Teaching Assistant** for CS6140 - **Machine Learning**, Khoury College, Northeastern University under Dr. Ehsan Elhamifar.
- **IEEE Peer Reviewer** - iSSSC 2025, INSTCon 2026.
- Winner of Honeywell's Industrial Automation Hackathon 2024 for designing a solution generating **20% productivity savings.**
- Winner at IEEE Computer Society's GirlGeekHack-2022 (**AI for Social Good**), selected from **100+** nationwide teams.